

RUMUS PERHITUNGAN PLTU 300MW

Konstanta Awal:

- Fuel Feed: 290 t/h
- Steam Valve: 85%
- Air Flow: 100%
- Water Inlet: 93%

TAHAP 1: FURNACE (Kimia menjadi Panas)

Hukum Fisika: *Hukum Kekekalan Massa & Termodinamika I*

Di sini, energi kimia batubara dilepaskan melalui proses oksidasi (pembakaran).

1 Konversi Laju Massa Batubara:

$$\dot{m}_{coal} = 290 \text{ t/h} \times \left(\frac{1000}{3600} \right) = 80.55 \text{ kg/s}$$

2 Kalkulasi Efisiensi Udara (AFR):

Karena *Air Flow* diatur 100%, kipas FD Fan menyuplai udara secara maksimal. Secara matematis di kode kita, ini tepat memenuhi rasio optimal pembakaran 4.5 kg udara per 1 kg batubara. Efisiensi AFR menjadi 1.0 (100%).

3 Total Panas Masuk (Q_{in}):

$$Q_{in} = \dot{m}_{coal} \times LHV \times \eta_{furnace} \times \eta_{AFR}$$

$$Q_{in} = 80.55 \times 17.5 \times 0.88 \times 1.0 = 1240.47 \text{ MW Thermal}$$

```
You, 4 days ago | 1 author (You)
class Furnace :
    def __init__ (self ):
        self .q_in_mw =0.0
        self .air_flow_system =AirFlowSystem ()

    def calculate_heat (self ,coal_feed_tph ,air_flow_pct ):
        mass_flow_kgs =coal_feed_tph *(1000.0 /3600.0 )
        eff_factor =self .air_flow_system .calculate_efficiency_factor (coal_feed_tph ,air_flow_pct )
        self .q_in_mw =mass_flow_kgs *config .LHV_COAL *config .FURNACE_EFF *eff_factor
        return self .q_in_mw
```

TAHAP 2: BOILER (Panas menjadi Tekanan Uap)

Hukum Fisika: *Hukum Termodinamika I & Persamaan Clausius-Clapeyron*

Di tahap ini, panas dari tungku memanaskan air menjadi uap. Kita asumsikan simulasi sudah mendekati titik stabil (*steady state*), di mana aliran air masuk hampir sama dengan uap yang keluar.

1 Keseimbangan Entalpi Absolut (Net Heat):

$$Q_{net} = Q_{in} + (\dot{m}_{air_masuk} \times h_{air}) - (\dot{m}_{uap_keluar} \times h_{uap})$$

Diketahui Entalpi Air (h_{air}) = 0.8 MJ/kg, dan Entalpi Uap (h_{uap}) = 3.5 MJ/kg.

2 Perhitungan Tekanan Otomatis:

Uap yang memuai akan menciptakan tekanan. Kode kita menggunakan turunan persamaan Clausius-Clapeyron untuk mencari tekanan berdasarkan suhu saat ini (533.3°C).

$$Exponent = 5423 \times \left(\frac{1}{540 + 273.15} - \frac{1}{533.3 + 273.15} \right) \approx -0.0554$$

$$P_{boiler} = 165 \times e^{-0.0554} \approx \mathbf{156.1 \text{ Bar}}$$

```
def update(self, q_in_mw, steam_flow_out_kgs, water_inlet_pct, dt=1.0):
    level_pct, feedwater_kgs = self.water_level.update(steam_flow_out_kgs, water_inlet_pct, dt)

    q_steam_out = steam_flow_out_kgs * config.ENTHALPY_STEAM_OUT
    q_water_in = feedwater_kgs * config.ENTHALPY_FEED_WATER

    net_heat_mw = q_in_mw + q_water_in - q_steam_out

    dt_temp = (net_heat_mw / (config.BOILER_MASS * config.CP_BOILER)) * dt
    self.temperature += dt_temp

    if self.temperature < 30.0:
        self.temperature = 30.0

    if self.temperature > 100.0:
        exponent = 5423.0 * ((1.0 / (config.NOMINAL_TEMP + 273.15)) - (1.0 / (self.temperature + 273.15)))
        try:
            self.pressure = config.NOMINAL_PRESSURE * math.exp(exponent)
        except OverflowError:
            self.pressure = 300.0
    else:
        self.pressure = 1.0

    return self.temperature, self.pressure, self.water_level.water_level_pct
```

TAHAP 3: STEAM VALVE (Laju Massa Uap)

Hukum Fisika: *Prinsip Bernoulli (Aliran Orifice)* Uap 156.1 Bar memaksa keluar melalui celah katup.

1 Menghitung Laju Massa Uap:

$$\dot{m}_{steam} = C_v \times \%_{buka} \times \sqrt{P_{boiler} - P_{turbine}}$$

Diketahui: $C_v = 45.0$, Bukaan = 0.85 (85%), Tekanan turbin konstan ≈ 5 Bar.

$$\dot{m}_{steam} = 45.0 \times 0.85 \times \sqrt{156.1 - 5}$$

$$\dot{m}_{steam} = 38.25 \times \sqrt{151.1} \approx 38.25 \times 12.29 = 470.18 \text{ kg/s}$$

```
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class Valve :
    def __init__(self):
        self.steam_flow_kgs = 0.0

    def calculate_flow(self, valve_opening_pct, p_boiler, p_turbine = 5.0):
        dp = max(0.0, p_boiler - p_turbine)

        compressibility_factor = 0.98

        self.steam_flow_kgs = (config.VALVE_CV * (valve_opening_pct / 100.0) * math.sqrt(dp) * compressibility_factor)
        return self.steam_flow_kgs
```

TAHAP 4: TURBINE (Dari Laju Uap menjadi Putaran Mekanik)

Hukum Fisika: *Ekspansi Isentropik Termodinamika II & Kurva Efisiensi Polinomial*

1 Faktor Deviasi (Polynomial Curve):

Karena suhu (533.3°C) dan tekanan (156.1 Bar) berada sedikit di bawah angka nominal (540°C & 165 Bar), entalpi uap sedikit berkurang. Kurva polinomial kita menghitung deviasinya:

$$\text{Deviasi} \approx +0.00009 \text{ (Faktor koreksi sangat presisi)}$$

$$\Delta H_{actual} = 0.8 \times (1.00009) \approx 0.8 \text{ MJ/kg}$$

2 Kalkulasi Daya Mekanik Poros (P_{mech}):

$$P_{mech} = \dot{m}_{steam} \times \Delta H_{actual} \times \eta_{turbine}$$

$$P_{mech} = 470.18 \times 0.8 \times 0.85 = 319.72 \text{ MW Mekanik}$$

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```
class Turbine:
    def __init__(self):
        self.p_mech_mw = 0.0

    def calculate_power(self, steam_flow_kgs, t_boiler, p_boiler):
        t_dev_curve = 0.015 * ((t_boiler - config.NOMINAL_TEMP) ** 2) / 10000.0
        p_dev_curve = 0.02 * ((p_boiler - config.NOMINAL_PRESSURE) ** 2) / 10000.0

        if t_boiler < config.NOMINAL_TEMP:
            t_dev_curve = -t_dev_curve

        delta_h_actual = config.ENTHALPY_DROP_NOMINAL * (1.0 + t_dev_curve + p_dev_curve)

        self.p_mech_mw = steam_flow_kgs * delta_h_actual * config.TURBINE_EFF

        if self.p_mech_mw < 0:
            self.p_mech_mw = 0.0

        return self.p_mech_mw
```

TAHAP 5: GENERATOR (Dari Mekanik menjadi Listrik)

Hukum Fisika: *Hukum Induksi Elektromagnetik Faraday*

$$P_{elec} = P_{mech} \times \eta_{generator}$$

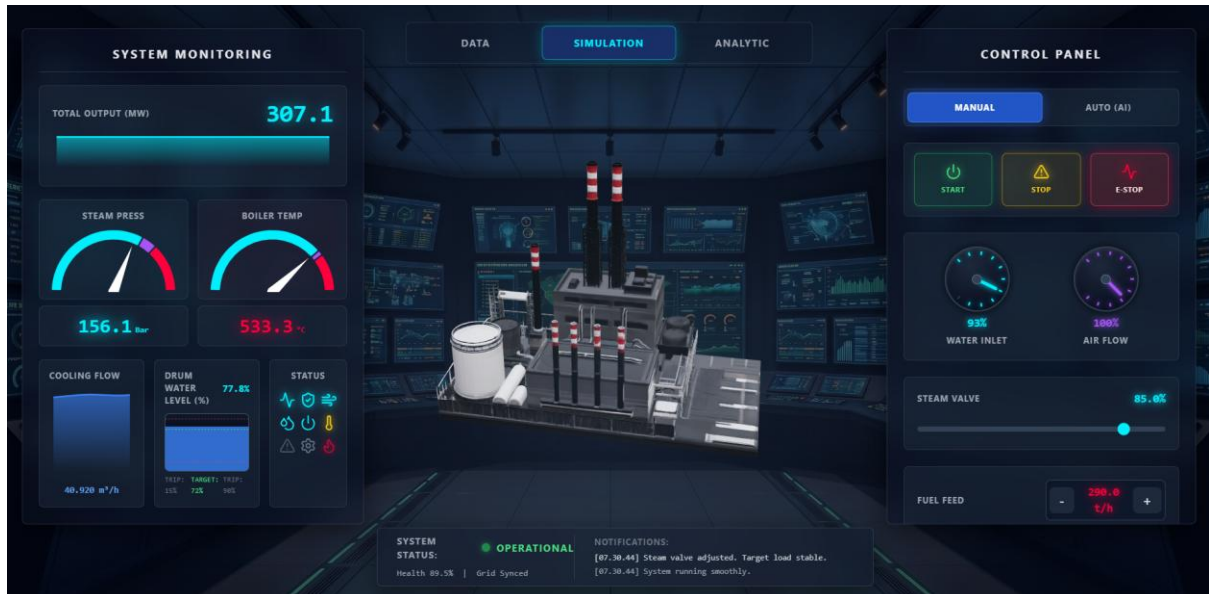
$$P_{elec} = 319.72 \times 0.98 = \mathbf{313.3 \text{ MW Listrik}}$$

You, 4 days ago | 1 author (You)

```
class Generator :
    def __init__ (self ):
        self .p_elec_mw =0.0

    def calculate_electrical_power (self ,p_mech_mw ):
        """
        Rumus: P_elec = P_mech * Efisiensi_Generator
        """
        self .p_elec_mw =p_mech_mw *config .GENERATOR_EFF
        return self .p_elec_mw
```

Hasil pada simulasi:



Terjadi selisih +6mw karena adanya toleransi panas (98%) yang tidak terkonversi sempurna menjadi listrik namun diserap oleh komponen PLTU